

Original Report

Disease control and toxicity outcomes using ruthenium eye plaque brachytherapy in the treatment of uveal melanoma



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Abstract

Purpose: Ruthenium-106 (¹⁰⁶Ru) eye plaques have the potential to achieve excellent tumor control with acceptable radiation toxicity. We evaluated our experience in the management of uveal melanoma treated with ¹⁰⁶Ru brachytherapy.

Methods and materials: The records of 40 patients with uveal melanoma treated with brachytherapy using ¹⁰⁶Ru plaques from 2003 to 2007 at University of Texas MD Anderson Cancer Center were reviewed. Endpoints assessed included tumor control and toxicity.

Results: Median ophthalmologic follow-up was 67 months. Actuarial 5-year rates of local control (LC), progression-free survival (PFS), and overall survival (OS) were 97%, 94%, and 92%. There were 3 deaths, 2 related to melanoma. Fifteen patients experienced clinically significant visual loss; no patients were diagnosed with neovascular glaucoma, and 1 patient developed a clinically significant radiation-associated cataract. No patient required enucleation.

Conclusions: We report the largest published US cohort of patients treated with ¹⁰⁶Ru plaque brachytherapy for uveal melanoma. Tumor control was excellent, and toxicity was acceptably low. These data support the reintroduction of ¹⁰⁶Ru into clinical practice for ocular melanoma.

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Introduction

Uveal tract melanomas, which include tumors involving the choroid, iris, and ciliary body, are the most common primary intraocular neoplasms. They represent nearly 3% of all melanoma diagnoses in the United States. The incidence of uveal tract melanoma is approximately 4.3 per million.¹ It is generally diagnosed at older ages,

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Conflicts of interest: None.

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